

Assessment 01 – Bioinformatics

Core Base

Why is Bioinformatics an Interdisciplinary Field?

Bioinformatics combines several scientific fields because biological data are too large and complex to study using traditional methods alone.

Biology

- Provides knowledge about genes, DNA, RNA, proteins, evolution, and diseases.
- Example: Studying mutations responsible for inherited diseases.

Computer Science

- Develops software, databases, and algorithms for storing and analyzing biological information.
- Example: Using sequence alignment software to compare genomes.

Mathematics

- Used in sequence alignment, evolutionary analysis, and prediction models.
- Example: Calculating similarity scores between DNA sequences.

Statistics

- Essential for analyzing gene expression and sequencing data.
- Example: Identifying genes that are significantly associated with a disease.

Overall idea: Bioinformatics succeeds because these disciplines work together to solve biological problems more efficiently.

Q1:

Scenario

A researcher has discovered a DNA sequence related to a rare genetic disorder and wants to know whether it has been reported before.

Recommended databases

GenBank

- One of the largest public DNA sequence databases.
- Stores nucleotide sequences submitted by researchers worldwide.
- Helps determine whether the DNA sequence has already been published.

NCBI Nucleotide Database

- Contains DNA and RNA sequences from multiple sources.
- Provides sequence information, annotations, and references.
- Useful for checking previous studies involving the sequence.

OMIM (Online Mendelian Inheritance in Man)

- Focuses on human genes and inherited diseases.
- Helps determine whether the sequence is linked to a known genetic disorder.

dbSNP (optional if discussed in class)

- Contains information about genetic variations.
- Useful for identifying disease-associated mutations.

Why these databases?

Together they allow the researcher to:

- check if the sequence already exists,
- identify related genes,
- find disease associations,

- review previous research before performing laboratory experiments.

Q2:

Scenario

Compare the genome of a disease-causing bacterium with related bacterial species.

Recommended databases

NCBI Genome Database

- Contains complete genome sequences of thousands of organisms.
- Allows comparison between bacterial genomes.

Ensembl Bacteria

- Provides annotated bacterial genomes.
- Includes gene locations and functional information.

KEGG

- Shows metabolic pathways and gene functions.
- Helps understand how conserved genes contribute to bacterial survival or disease.

Why these databases?

They provide:

- complete genome sequences,
- gene annotations,
- evolutionary relationships,
- functional information needed to identify conserved genes shared among related bacteria.

Q3:

Scenario

A biotechnology company has isolated a protein and wants to know whether similar proteins exist in other organisms.

Recommended databases

UniProt

- The most comprehensive protein database.
- Provides protein names, functions, sequences, domains, and biological roles.

Protein Data Bank (PDB)

- Contains three-dimensional protein structures.
- Helps study protein shape and possible function.

NCBI Protein Database

- Stores protein sequences from many organisms.
- Useful for finding similar proteins across species.

Why these databases?

The researcher can learn:

- whether similar proteins already exist,
- their biological functions,
- evolutionary relationships,
- structural information,
- possible industrial or medical applications.

Q4:

Scenario

A scientist has an unknown protein sequence and wants to determine its function.

Recommended databases

UniProt

- First choice for protein function.
- Includes experimentally verified protein annotations.

Pfam

- Identifies conserved protein domains.
- Domains often indicate protein function.

InterPro

- Combines information from several protein family databases.
- Predicts biological function based on conserved motifs.

Protein Data Bank (PDB)

- If a similar protein structure exists, structural similarity can suggest function.

Why these databases?

Combining sequence, conserved domains, protein families, and structural information gives a much more reliable prediction of protein function than using a single database.

Q5:

Scenario

You receive an unknown biological sequence and do not know whether it is DNA, RNA, or protein.

Step-by-step approach

Step 1: Examine the sequence

- Check which letters are present.
- DNA contains A, T, C, G.
- RNA contains A, U, C, G.
- Protein sequences include 20 amino acid letters.

Step 2: Search nucleotide databases

If the sequence appears to be DNA or RNA:

- Search the NCBI Nucleotide Database or GenBank.
- Determine whether identical or similar sequences exist.

Step 3: Search protein databases

If it appears to be a protein:

- Search UniProt or the NCBI Protein Database.
- Look for similar proteins and functional annotations.

Step 4: Perform sequence similarity analysis

Use BLAST.

- Compare the sequence against known biological databases.
- Identify similar sequences and possible species.

Step 5: Study functional information

If protein matches are found:

- Check UniProt, Pfam, InterPro, or PDB.
- Investigate protein domains, families, and biological functions.

Step 6: Draw conclusions

Combine evidence from sequence similarity, annotations, conserved domains, and literature to identify whether the unknown sequence is DNA, RNA, or protein and predict its biological role.

